

Technical Note

Fractal analysis of dendrite morphology using modified box-counting method

Dušan Ristanović^{a,*}, Bratislav D. Stefanović^b, Nela Puškaš^b^a Department of Biophysics, Medical Faculty, University of Belgrade, Serbia^b Department of Histology, Medical Faculty, University of Belgrade, Serbia

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ABSTRACT

The box-counting dimension of a non-stellate neuron changes continuously with its rotation. During preprocessing for box-counting, non-stellate neurons should be arranged so that the major diameters of their dendrite fields are parallel. A non-stellate neuronal picture should have the smallest fractal dimension when the angle between the horizontal axis and its major diameter is about 45°. The box-counting method does not consider the position of a picture on the computer screen. Therefore a dispersion of the box dimension values of a neuronal sample is rather large and their mean value is with larger variance. Modified box-counting method partly diminishes these findings. To improve a dependence of neuronal rotation on the box-counting dimension of non-stellate neurons, prior to applying box-counting method, non-stellate neurons should be arranged so that the major diameters of their dendrite fields are parallel.

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1. Introduction

It is well-known that a picture can exhibit rotational symmetry if rotation by a specific angle about some central axis point can return the picture to its original configuration. Squares and hexagonal snowflakes are familiar examples: rotation by 90° of a square or by 60° of a hexagonal snowflake about their centers returns the shapes to their original configurations. It should be reasonable to presume that the fractal (box) dimensions (FDs) of such a picture in these positions have the same values. Neurons with dendrites homogeneously filling a spherical volume are termed *stellate* (Fiala and Harris, 2002). Dendrites of such neurons radiate in all directions from the cell body. A projection of a stellate neuron on a plane gives rise to a picture in which projections of dendrites also radiate in all directions from the cell body projection within a circle. In contrast, neurons lacking strong radial symmetry can be termed *non-stellate*. One of the aims of the present paper is to explore how the FD changes in the course of continuous rotation of non-stellate neurons. We propose a modification to box-counting methods to address rotational variation in non-stellate neurons. We recommend arranging pictures of non-stellate neurons of a given sample so that, when possible, the major diameters of the neuronal fields are parallel, prior to applying box-counting methods. Our

modification of the box-counting method helps to diminish possible errors caused by rotation.

2. Materials and methods

2.1. Experimental

Male Wistar rats were used in this study. A detailed description of the histological procedure used and data processing can be found in Ristanović et al. (2013). We considered a sample of 16 cortical pyramidal (non-stellate) neurons (see Fig. 1, where the pictures of neurons are arranged vertically using rotation). The drawings were analyzed after removing cell bodies, spines and basal dendrites. We used tools available in the public domain software ImageJ (<http://rsbweb.nih.gov/ij>) to process the images to skeletonized formats and calculate the FDs.

2.2. Box-counting method which uses ImageJ

By default, the box sizes in the table for counting the FD using ImageJ software are 2, 3, 4, 6, 8, 12, 16, 32, 64 and the FD for Neuron 1 (Fig. 1) is 1.2741. In this paper we used an increasing geometric progression 2^n where $n = 0, 1, 2, \dots, 10$ or more, and found $D = 1.2255$. It means that the FD depends on the number and kind of box sizes. Therefore, for calculating SDs in a sample it is necessary to use the same series of sizes. The graph in the table is wrong since the physical property, as length, does not have the logarithm. The measured

* Corresponding author. Tel.: +381 11 3607061; fax: +381 11 3607061.
E-mail address: dusan@ristanovic.com (D. Ristanović).

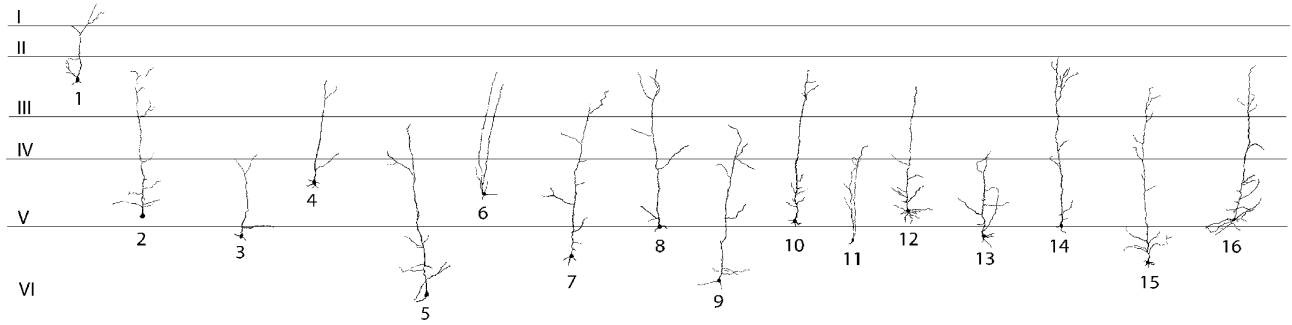


Fig. 1. Sixteen representative examples of pyramidal neurons. Neurons are selected within rat's frontoparietal cortex being dorsally and dorsolaterally positioned in the coronal sections of the rat forebrain and of cerebral cortex hemispheres. The cortical layers are denoted by Roman numerals. Scale bar: 50 μm .

data should be directly inserted in the coordinate system with logarithmic axes. Finally, straight line segments have the FDs less than 1.

2.3. Statistics

In the case of small samples, where the number of variates is 30 or less (Alder and Roessler, 1972), we have adopted the criteria for normality of data distribution offered by Kurickiy (1969). The details of the method can be seen in Ristanović et al. (2009). If

$$\left| \frac{\sigma_3}{A} \right| \leq 2.0 \quad \left| \frac{\sigma_4}{E} \right| \leq 2.0 \quad (1)$$

where σ_3 and σ_4 are the mean standard deviations of data sets related to the *asymmetry* A and *excess* E , respectively, there are grounds for accepting that the data are normally distributed.

For a set of n variates, $X_1, X_2, X_3, \dots, X_n$, whose mean is X_{av} , the best estimate of the standard deviation of the population, denoted by s , is given by

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - X_{av})^2}{n - 1}} \quad (2)$$

The square of the standard deviation is called the *variance*. The variance is widely used as a measure for dispersion of a set of variates about their mean. If X and Y are normally distributed populations with identical standard deviations, then the ratio of the squares of the best estimates of the standard deviations (the ratio of the variances), that is,

$$F = \frac{s_x^2}{s_y^2} \quad (3)$$

where

$$s_x^2 = \frac{\sum_{i=1}^{n_x} (X_i - X_{av})^2}{n_x - 1} \quad s_y^2 = \frac{\sum_{i=1}^{n_y} (Y_i - Y_{av})^2}{n_y - 1} \quad (4)$$

satisfy an F -distribution with $n_x - 1$ and $n_y - 1$ degrees of freedom.

3. Results

3.1. Elementary geometry of screen's pixels

In a skeletonized (one-pixel-wide) line on a screen two adjacent pixels can be in touch either by their sides or by their vertices (Ristanović and Milošević, 2012). In the first case small pixels can form horizontal or vertical straight lines, while in the second case they form an oblique straight line so that the angle from the horizontal to oblique line is 45° (Fig. 2). From this figure it is seen that the same geometrical length (r) can be achieved either with 10 horizontally or vertically arranged squares (pixels), or with approximately 7 squares arranged as an oblique straight line.

To illustrate this finding we present two examples, as follow:

- (1) Consider a skeletonized square (side length 60 mm) drawn on the computer screen at a resolution of 300 ppi. From its starting position (two opposite sides of the square are horizontal and other two are vertical) denoted by 0, the square rotates about its center for a series of angles $7.5^\circ, 15^\circ, 22.5^\circ, 30^\circ$, etc. to 90° . From Fig. 3A it is seen that the box dimension of the square decreases when the angle of the square changes from 0° to 45° . After that the box dimension increases. In Fig. 3B where the square is considered as a binary image (the width of the lines is 1 pt) a similar result is obtained.
- (2) From Fig. 1 we selected, at random, one neuron (say, neuron 5), chose its apical dendrite and performed with it the same procedure as with the square. In Fig. 4 it is shown that the FD of the neuron also reaches minimum at the value around 45° .

3.2. Testing the samples for normality

To apply parametric statistics to our data, it is necessary to show that populations involved are normally distributed (Alder and Roessler, 1972). For that purpose, we selected two samples from these populations: one sample was a group of 16 neurons which are vertically oriented (Fig. 1), and the other sample was the same group of neurons but rotated for 45° . Using the method shown in Section 2.3 we show that both samples mentioned are normally distributed since the absolute values of σ_3/A and σ_4/E are not larger than 2.0 (Table 1).

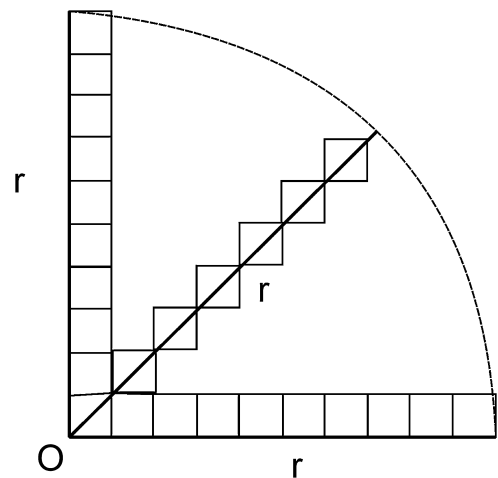


Fig. 2. Geometry of screen pixels. The length r of the series of horizontal and vertical pixels (10 squares) is approximately equal to the length r of oblique pixels (7 squares). O is the center of a circle (shown by broken line) of the radius r .

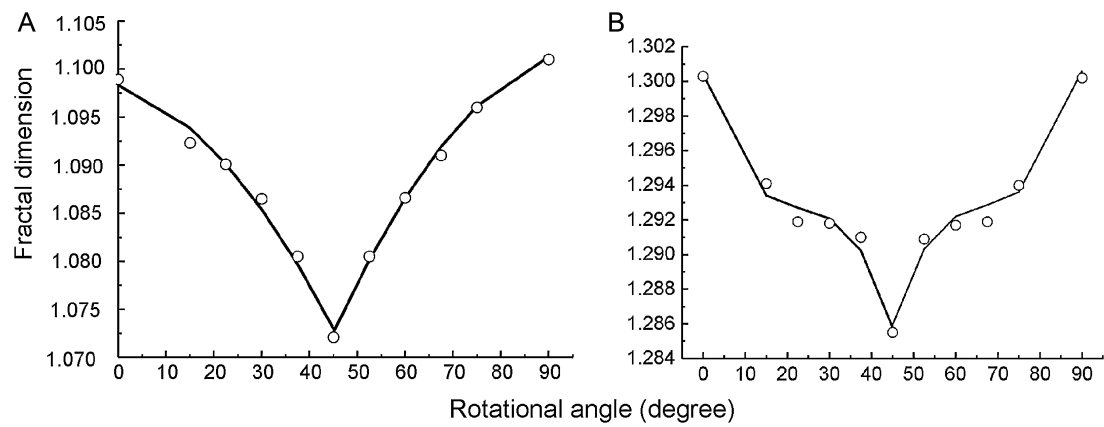


Fig. 3. Relations between the fractal dimensions and the rotational angle for a square. (A) Relation for the skeletonized square. (B) Relation for the binary square.

Table 1
Results of testing for normality of the box dimensions of binary pyramidal cells. The neurons were in vertical position 0°, rotated for 45° and rotated between 20° and 320°. The parameters σ_3 and σ_4 are the mean standard deviations of asymmetry (A) and excess (E).

σ_3	σ_4	σ_3/A			σ_4/E		
		0°	45°	20°–320°	0°	45°	20°–320°
0.5279	0.8823	0.6	0.8	0.7	0.8	0.4	1.4

Table 2
Results of statistical analysis of the distribution of box-counting fractal dimensions for binary pictures of pyramidal neurons. The values of mean fractal dimensions are given as the means (D_{av}) and standard errors (SE) when the neurons were in vertical position 0° (Fig. 1), rotated for 45° and rotated between 20° and 320° (see text). Variable t is the Student's statistic and F is the Fisher's statistic.

$D_{av} \pm SE$			t		F
0°	45°	20°–320°	0°–45°	0°–(20°–320°)	
1.357 ± 0.005	1.325 ± 0.006	1.338 ± 0.008	4.301**	3.684*	2.63

* $p < 0.01$.
** $p < 0.001$.

3.3. Analysis of the paired experiment

The paired Student's t test was used to compare the means of two populations when each variate of one population was related to a corresponding variate of the other population (Dodge, 2008). We found that the mean FD of vertical neurons was significantly larger than the mean FD of the corresponding oblique neurons (Table 2).

3.4. The modified box-counting method

In order to make disorder of the neuronal pictures in Fig. 1, we rotated each neuronal picture so that the neuron 1 was rotated

for 20° from the starting (vertical) position, neuron 2 for 40°, neuron 3 for 60°, etc. and neuron 16 for 320°. First of all, we showed that the FD of such neurons were normally distributed (Table 1) and their means were significantly different from the means of the FDs of vertical neurons (t -test, Table 2). We applied the F -test and showed that the variances of disordered neurons were significantly larger than those of the ordered (vertical) neurons in Fig. 1 (Table 2) since the calculated F -values (Table 2) exceed the critical value $F_{.95}(15,15)=2.40$ for $n-1=15$ degrees of freedom. Assuming 5% level of significance we conclude that the two variances are significantly different.

The modified box-counting method is a recommendation to arrange pictures of neurons of a given sample with non-stellate dendrite fields so that the major diameters (Neale et al., 1993) of the fields are parallel, and after that preprocessing apply the box-counting method.

4. Discussion

Every scientist who researches using measurement knows that the result of a measurement does not worth much without the corresponding experimental error. Therefore every scientist chooses a method of measurement which enables the smallest experimental error. The rotation of neurons has been previously used and discussed by some authors (Montague and Friedlander, 1989, 1991; Fernández and Jelinek, 2001; Karperien et al., 2013). For example, Montague and Friedlander (1989) randomly varied the position and orientation of the overlay drawing relative to the grid. At each particular box size, this procedure was done 10 times. The number

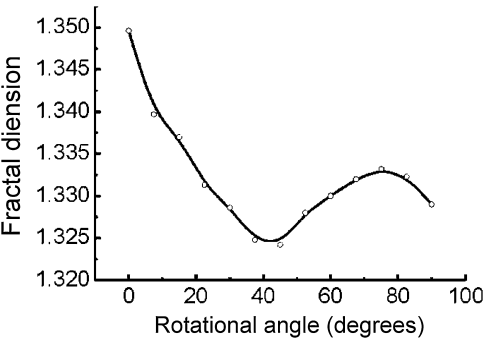


Fig. 4. Relations between the fractal dimensions and rotation angle for a binary neuron selected at random from Fig. 1. The minimum is about 40°. Data points are connected using the spline-B function.

of boxes intersected was then averaged. Thus, they calculated the mean FD for a neuron, while we calculated this mean but for a sample of neurons.

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